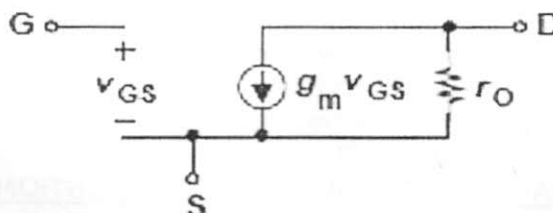
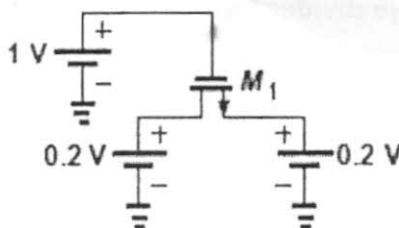


9. Saturation region of N-MOSFET is defined by following voltage conditions
- A) $V_{GS} > V_{TH}$, and $V_{DS} \geq V_{GS} - V_{TH}$ B) $V_{GS} < V_{TH}$
 C) $V_{GS} > V_{TH}$, and $V_{DS} < V_{GS} - V_{TH}$ D) $V_{GS} < V_{TH}$, and $V_{DS} < V_{GS} - V_{TH}$
10. For the MOSFET operating in saturation, if the channel width is doubled, drain current i_D changes by a factor
- A) 0.5 B) 4
 C) 2 D) No change
11. Which of the parameter is used to represent channel-length modulation in small signal model of MOSFET shown



- A) g_m B) r_o
 C) v_{GS} D) μ_n
12. If a MOSFET is to be used in the making of an amplifier then it must work in
- A) Cut-off region B) Triode region
 C) Saturation region D) Both cut-off and triode region can be used
13. When V_{DS} is increased, pinch off point of the MOSFET move
- A) Towards the source region B) Towards drain region
 C) Towards bottom of bulk D) Away from source region
14. Determine the region of operation of M1 in each of the circuits shown in Fig. Assume $V_{TH}=0.4V$



- A) Cut-off region B) Triode region
 C) Saturation region D) Sub threshold conduction
15. Channel between source and drain of a MOSFET acts as a
- A) Variable capacitance B) Variable inductance
 C) Variable resistance D) None of the mentioned
16. Sub threshold conduction of MOSFET means
- A) Conduction of MOSFET for gate voltages below the threshold voltage B) Saturation of velocity of charge carriers at high electric field
 C) Increase of charge carriers in depletion region when body voltage decreases D) The variation in drain current due to change in channel length caused by movement of Pinch-off point
17. The type of power amplifier which exhibits crossover distortion in its output is
- A) Class A B) Class B
 C) Class C D) Class AB
18. Which of the following statement is not correct for differential amplifier
- A) Amplifies the difference between two input signals B) Rejects the common mode signals
 C) Output voltage is given by $V_o = A_d V_d + A_c V_c$ D) Ideal value of CMRR for differential amplifier is 0
19. If output is measured between two Drain terminals of MOSFETs, then the Differential amplifier with two input signal is said to be configured as
- A) Dual Input Balanced Output B) Dual Input Unbalanced Output
 C) Single Input Balanced Output D) Single Input Unbalanced Output

20. What is cross over distortion?

- A) Distortion occurred due to resistors
C) Distortion occurred due to inductors

- B) Distortion occurred due to Capacitors
D) Effect occurred during switching of transistor after every half cycle

PART - B: DESCRIPTIVE ANSWER QUESTIONS**Unit - I**

1. a) Write the circuit diagram of BJT with collector feedback bias. Explain the DC analysis for the same.
b) Given the information appearing in Figure. Q1(a), determine:
i. I_C ii. V_E iii. V_{CC} iv. V_{CE} v. V_B vi. R_1 . Assume $V_{BE}=0.7V$.

Marks	BT*	CO*	PO*
8	L*2	1	1

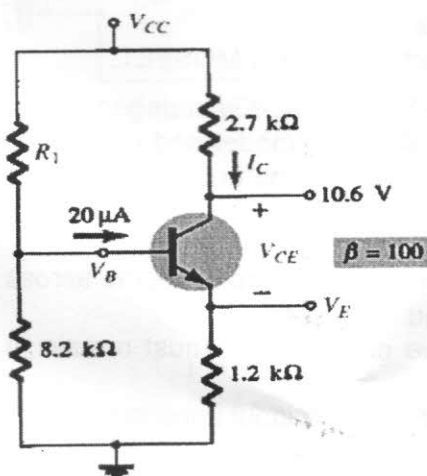


Figure. Q1(a)

2. a) For emitter unbypassed CE emitter bias circuit, derive the expression for the input impedance, output impedance and voltage gain using r_e model of transistor. Neglect the effect of r_o .
b) Design an CE RC coupled amplifier(Exclude the design of coupling and bypass capacitors) for following specifications
i. Supply voltage of 12V and collector current of 2mA.
ii. Magnitude of gain $A_v \geq 50$
Assume $\beta = 110$, $V_{CE} = 50\%$ of V_{CC} , $V_E = 10\%$ of V_{CC}

8	L3	1	1
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8	L2	2	1
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8	L3	2	1
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3. a) For Emitter follower configuration, draw the small signal circuit. Derive the expression for input impedance, output impedance and voltage gain neglecting Early Effect.
b) Given that $I_{CQ} = 2$ mA and $V_{CEQ} = 10$ V, determine R_1 and R_C for the network of Fig. Q3(b).

8	L2	2	1
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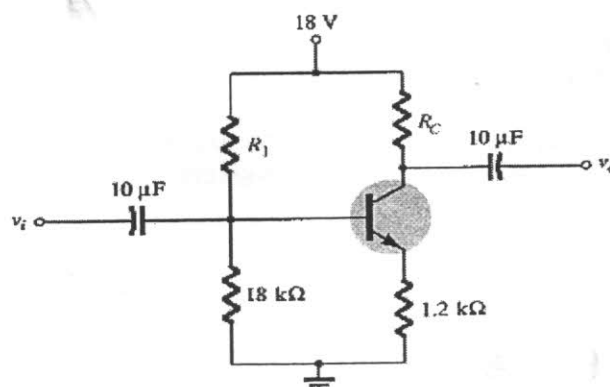


Figure. Q3(b)

8	L3	1	1
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Unit – II

4. a) Determine the bias current of M_1 in Fig. Q4(a) assuming $V_{TH} = 0.5V$, $\mu_n C_{ox} = 100 \mu A/V^2$, $W/L = 5/0.18$, and $\lambda = 0$. What is the maximum allowable value of R_D for M_1 to remain in saturation?

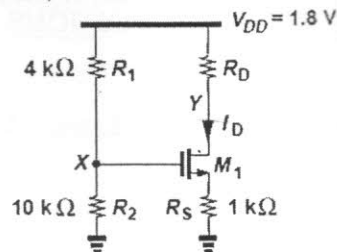


Figure. Q4(a)

- c) With relevant plot explain the output characteristics of N-MOSFET.
5. a) Draw a neat circuit diagram of a Common Source stage with degeneration and voltage divider biasing. Draw its small signal model and obtain the expression for small signal gain, input impedance and output impedance considering $\lambda=0$.
- b) The circuit in Fig. Q5b must be designed for a voltage drop of $200mV$ across R_S . Assuming $\mu_n C_{ox} = 200 \mu A/V^2$, $\lambda = 0$, and $V_{TH} = 0.4V$,
 (i) Calculate the minimum allowable value of W/L if M_1 must remain in saturation
 (ii) What are the required values of R_1 and R_2 if the input impedance must be at least $30k\Omega$

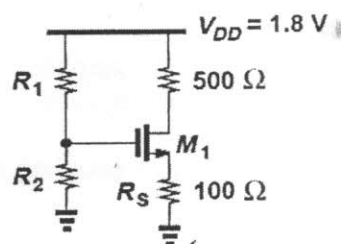


Figure. Q5(b)

6. a) Draw a neat circuit diagram of a Common Source stage with degeneration. Draw its small signal model and obtain the expression for small signal gain considering $\lambda \neq 0$.
- b) Explain the following with respect to MOSFET
- Body effect
 - Velocity saturation
 - Subthreshold-Conduction
 - Channel Length Modulation

Unit – III

7. a) Draw the circuit diagram of MOS differential pair. Derive the expression for small signal differential gain
- b) Draw the circuit diagram of NMOS current mirror and derive the expression for I_{copy} . Calculate I_{copy} in the circuit shown in Figure. Q7(b). Assume all of the transistors operate in saturation.

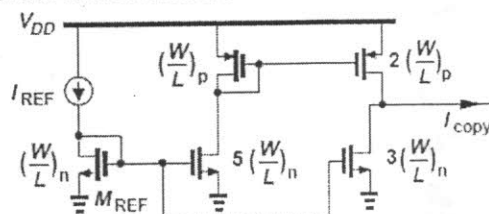


Figure. Q7(b)

- 8 L2 5 1



Figure. Q8(b)
BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
